SIMATS ENGINEERING

Lab Programs

COURSE CODE: CSA1418 **COURSE NAME: Compiler Design**

Experiment 9: C Program to Eliminate Left Recursion from a Given CFG

Aim:

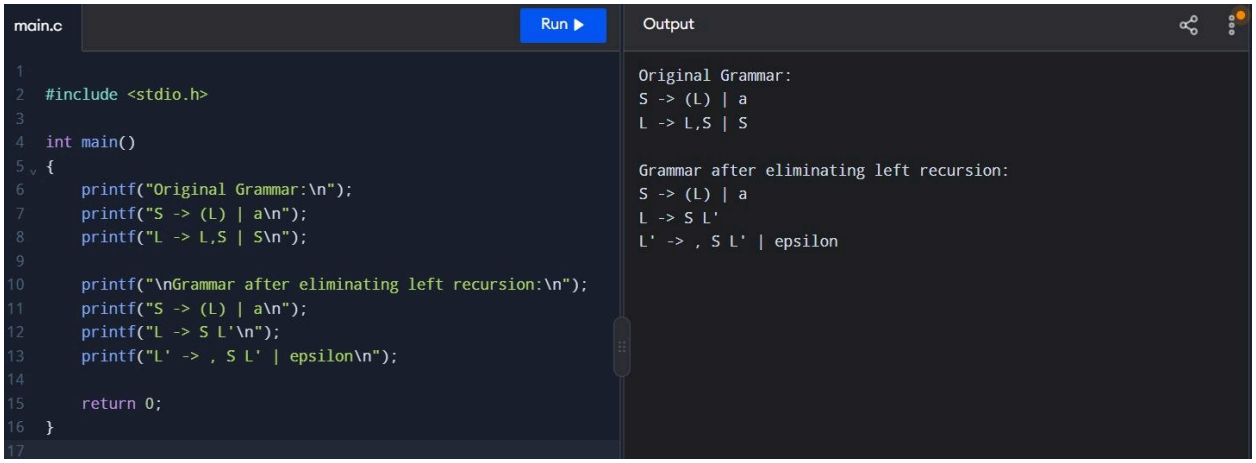
To write a C program to eliminate left recursion from the given context-free grammar (CFG).

Software Used: Programiz Online C Compiler.

Algorithm / Procedure:

- Open the Programiz Online C Compiler.
- Write the C program to display the grammar before and after eliminating left recursion.
- Consider the grammar:
    - S → (L) | a
    - L → L,S | S
- Identify the immediate left recursion in the production L → L,S | S.
- Apply the left-recursion elimination technique.
- Transform the productions into L → SL′ and L′ → ,SL′ | ε.
- Display the original and transformed grammars.
- Compile and execute the program.
- Verify the output.

Execution Screenshot:



Result:

Thus, the C program was implemented and executed successfully to display the grammar after eliminating left recursion.